What is a variable?

Variable is a container use to store data and value

i) Var :-

Value can changed it has function scope.

ii) let :-

Value can be changed it has block scope.

iii) Const :-

Value cannot be reassigned and it has block scope.

Example :-

```
var age = 20;
age = 21;

let mark = 50;
mark = 60;

Const Pi = 3.14;
// Pi = 4;
```

2) Java Script Data types:

Data type is a what kind of value we are storing.

i) String -> text

let name = "vijay";

ii) Number -> numbers

let age = 20;

iii) Boolean -> true or false

let isstudent = true;

iv) undefined -> value is not assigned

let x;

v) null -> intentionally empty value

let y = null;

---

3) What is the different between;

i) == is called loose quality

it check the values and can convert its type.

eg: 5 == "5"

output: true.

ii) === is called strict equality:-

It check value + data type

Eg: 5 === "5"

Output: false

---

4) Explain these operators with examples:-

i) Arithmetic operators:-

* Arithmetic operators use for mathmatical Calculation

* +, -, *, /, % They are operator symbols.

Example:-

```
let a = 10;
let b = 5;
Consol.log(a+b);
Consol.log(a-b);
```

ii) Assignment operators:-

* Assignment operators used to assign / values update

* =, +=, -=, *=, /=

Example:

```
let x = 10;
x += 5;
```

OR x = 5

iii) Comparison operators:-

* comparison operators used to compare values.

* Symbols is >, <, >=, <=, ==, ===, !=

Example:-

10 > 5

Output:-

True.

iv) Logical operators:-

* used to combine conditions

- Symbol &&, ||, !
- && -> AND
- || -> OR
- ! -> NOT

Example:-

age > 18 && age < 30

output:-

Both conditions be true.

---

5) what is the difference between:-

Array:-

Array -> Store multiple value in one variables

Eg:- let fruits = ["Apple", "mango", "Orange"];

Object :-

Object stores data using key - value pairs

Eg:-

```
let student ={
    name : "vijay";
    age : 21
};
```

Array of object :-

Multiple objects stored inside an array

Eg:-

```
let student =[
    { name : "Shalu ;", age : 19 },
    { name : "vijay ;", age : 21 }
];
```

---

6) what is functions :-

A function is a reusable block of code Reforms a specific task

Eg:-

```
function add ( a,b ) {
    return a+b ;
}
```

Parameter :-

The variables written inside the function definition.

Argument :-

The actual value passed when calling the function

Return :-

Send a result back from the function.

---

7) EVEN or ODD :-

```
let number = 28;
if (number % 2 === 0) {
    console.log("Even");
}
else {
    console.log("odd");
}
```

Output :- 27 is odd.

8) using for loop:-

```
for (let i=1; i<=10; i++){
    if (i%2 ===0){
        Console.log(i);
    }
}
```

Output:-

2, 4, 6, 8, 10

9) factorial:-

```
let number =5;
let factorial =1;
for (let i=1; i<= number; i++){
}
Console.log(number + "! = " + factorial);
```

Output:-

5! = 120

10) Fibonacci Series

```
let a=0;
let b= 1;
for (let i = 0;  i< 8;  i++){
    Console . log(a);
   let nesct = a+b;
   a= b;
   b= nesct;
}
```

Output

0
1
1
2
3
5
8
13

---

11) Array :-

```
let numbers = [10, 15, 20, 25, 30, 35, 40];
for (let i=0; i< numbers. length ; i++){
   if numbr [i] % 2 ===0){
      Console .log (numbr [i]);
   }
}
```

Output:-

10
20
30
40

12) Array total:-

```
let marks = [80, 75, 90, 85, 70];
let total = 0;
for (let i = 0; i < mark.length; i++){
    total += total + marks[i];
}
Consoh.log("total = " + total);
```

Output:-

total = 400

13) Object :-

```
let student = {
    name = "vijay",
    age : 21;
    Course : "Full stack",
    mark = 85
};
```

```
Console.log("student name : " + student.name);
Console.log("course : " + student.course);
Console.log("mark : " + student.mark);
```

output :

Student : vijay

Course : full stack

mark : 85

---

14) Array of objects :

```
let students = [
    { name : "Shalu", mark : 80 },
    { name : "Babu", mark : 45 },
    { name : "vijay", mark : 90 },
    { name : "Kumar" mark : 35 },
];

for (let i = 0; i < student.length; i++) {
    if (students[i].mark >= 50) {
        Console.log(students[i].name + "_" + students
                                        [i].mark);
    }
}
```

Output:

Vijay - 90

Shalu - 80

---

19) function - Largest number

```
function find Largest (a, b){
    if (a > b) {
        return a;
    }
    else {
        return b;
    }
}
console.log ( find Largest (10, 20));
```

Output:

20
